

Week 6

Evaluating Data Quality and Identifying Bias

Certificate IV in Information Technology

Data Science and AI

East Perth (Face to Face)

Learning Objectives

- Recognize key dimensions of data quality, including timeliness, authority, relevance, and completeness.
- Identify common sources of bias in large datasets.
- Apply basic techniques to evaluate data quality and detect bias.
- Explain how data quality and bias influence machine learning and AI outcomes.
- Build practical skills that align with industry standards and workplace expectations.

Why Data Quality and Bias Matter

- The accuracy and fairness of AI systems depend on the quality and neutrality of the data used for training.
- Poor data can lead to flawed decisions, unreliable AI behavior, and unintended social or business consequences.
- Addressing data quality and bias reduces risk in AI projects and improves outcomes for organizations and users.

Dimensions of Data Quality

Dimension	Description
Timeliness	Data is up to date and reflects current reality.
Authority	Source is credible and reliable.
Relevance	Data is applicable to the problem or question.
Accuracy	Data is correct and free of errors.
Completeness	All necessary data fields and records are present.
Consistency	Data is formatted and recorded uniformly.

Practical Example

Evaluating Data Quality

- High-quality data example; A patient medical record with up-to-date, verified, and complete details.
- Poor-quality data example; An online form missing contact details, with inconsistent spellings and outdated information.
- Discussion; Which real-world problems could be caused by poor data in these cases?

Introduction to Bias in Datasets

- Bias means systematic errors or imbalances in data that lead to unfair or unrepresentative outcomes.
- Main sources; Sampling bias (who or what is included), measurement bias (how data is collected), labeling bias (how data is categorized), and historical/social bias.
- In large AI datasets, even small biases can scale up to have major effects.

Industry Example

How Bias Influences AI Outcomes

- A facial recognition system was found to misidentify people with darker skin at much higher rates due to imbalanced training data.
- Language models can reproduce or amplify stereotypes if their training text is not representative of all groups.
- These failures can lead to public mistrust, legal risks, and operational problems for organizations.

Methods for Detecting Bias and Assessing Data Quality

- Exploratory Data Analysis (EDA) helps find missing data, outliers, and unbalanced classes.
- Visual tools like histograms and boxplots reveal patterns or gaps in representation.
- Compare dataset demographics with real-world or target population statistics.
- Documentation review; Check source notes, datasheets, or data cards for warnings.
- Use data profiling tools in Python (pandas profiling, seaborn visualizations, HuggingFace Datasets utilities).

Activity

Spot the Issues in a Real Dataset

- Work with a sample UCI dataset in small groups.
 - Step 1; Check for missing data and inconsistent values using Python or manually.
 - Step 2; Examine how classes or categories are distributed; is anyone or anything underrepresented?
 - Step 3; Document one example of poor data quality and one example of possible bias.
- Share findings with the class; discuss solutions and mitigation strategies.

Data Quality and Bias in AI Workflows

- Industry relies on documentation standards like "datasheets for datasets" to identify and record data quality and bias concerns.
- Previous week; You explored documentation practices; now, use this knowledge to evaluate real datasets.
- Effective mitigation includes data cleaning, re-sampling, clearer documentation, and stakeholder consultation.
- Transparency and reproducibility are essential for responsible AI development in the workplace.

From Classroom to Workplace

Preparation for Assessment and AI Projects

- Skills in evaluating quality and bias are core for industry roles; these are expected in AI research, data engineering, machine learning, and compliance jobs.
- Next; Your assessment tasks will require you to critically evaluate a large open dataset and report on documentation and ethical considerations.
- Practice these techniques now to prepare for Weeks 10-14 projects.

Summary and Next Steps

- Data quality and bias are foundational for trustworthy, safe, and effective AI outcomes.
- You now have practical tools to start evaluating and improving datasets for real-world projects.
- Next week; Learn core techniques for data cleaning and preprocessing.
- Self-check; Can you
 - Name three dimensions of data quality?
 - Give an example of bias and how it might impact AI?
 - Suggest one way to detect bias in a real dataset?

